

Breathing and circulation

Not all animals have lungs, but they all need oxygen. The cells that make up their bodies use oxygen in chemical processes that release energy from food and keep them alive. At the same time, an animal gets rid of carbon dioxide, which its cells produce as waste. For some animals, these two tasks are simple, because gases seep, or diffuse, directly through their skin. But most animals do not work like this. They breathe through lungs or gills, and they use blood to circulate oxygen and nutrients to their cells and to carry away waste. Blood is pumped by the heart. A glass frog's heart weighs less than 0.04 oz (1 g), but a blue whale's can be more than half a ton.



Oxygen from water

Like most animals that live in water, the axolotl draws oxygen out of the water through its gills. Gills are flaps or feathery outgrowths that have a rich blood supply. When water flows over them, oxygen passes into the animal's blood, and carbon dioxide flows out. The axolotl has external gills, which spread out just behind its head. Most fish have internal gills in the cavity behind their mouths. In addition to being used for breathing, gills sometimes double up as filters, helping animals to collect their food.

Thin skin lets oxygen and carbon dioxide spread into and out of the body

Life in a bubble

All spiders get their oxygen from the air, through lungs on their undersides. These lungs are called book lungs, because they are made up of tiny flaps like the pages of a book. But one spider, the water spider, spends most of its life beneath the surface of ponds. It makes an air-filled "diving bell," which it holds under water with a net of silk. When the spider hunts, it carries a slim film of air, which works like an aqualung, on the rear of its body, or abdomen.

Air collected by the spider on frequent visits to the pond surface



Bodies that breathe

Like shadowy ghosts, sea butterflies swim through the oceans using transparent flaps that look like wings. These tiny mollusks are snails, but they do not have shells, and they get all the oxygen they need without gills or lungs. Because their bodies are flat and thin, oxygen diffuses into them from the water, while carbon dioxide diffuses out. This way of exchanging gases is used by many small animals that live in water and also by earthworms on land.